# 19. Time-Varying Treatments and G-Methods

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Time-Varying Treatments, Treatment-Confounder Feedback, and G-Methods* (Prof. Sanghack Lee, GSDS SNU)

The central question is how to estimate the effect of a sequence of treatment decisions when covariates measured along the way are both affected by earlier treatment and drive later treatment, a situation in which ordinary regression, stratification and matching all fail.

These notes consolidate the slides and the appendix proofs: the identification assumptions, the g-formula and its iterated-conditional-expectation implementation, inverse probability weighting with (stabilized) weight products and marginal structural models, g-estimation of structural nested models, doubly robust estimation, dynamic strategies, positivity, and a comparison of the methods. Physical PDF page numbers are listed in Section 8. References: Hernán and Robins (2020), *What If*, Ch. 19–21; Naimi, Cole and Kennedy (2017).

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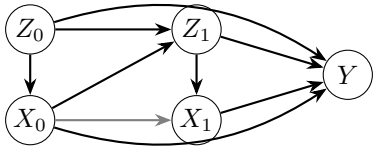
# 1 Time-varying treatments

**Purpose.** Set up sequential decisions and explain why feedback breaks standard adjustment.

**Recall the single time point.** Identification by back-door adjustment, $\mathbb{E}[Y(x)] = \sum_z \mathbb{E}[Y \mid X = x, Z = z]P(Z = z)$, and estimation by IPW, $\mathbb{E}[Y(x)] = \mathbb{E}[\mathbb{1}(X = x)Y/P(X = x \mid Z)]$. What if treatment is applied repeatedly?

Many interventions involve repeated decisions: medication updated at each visit, policies applied in stages, algorithms adapting to the current state. Later decisions are made after earlier information has been observed, so the data-generating process is history-dependent: action $\to$ state $\to$ action $\to \cdots \to$ outcome.

**Causal diagrams.** Treatments $X_k$ with history $\bar{X}_k = (X_0, \ldots, X_k)$; covariates (states) $Z_k$ with history $\bar{Z}_k = (Z_0, \ldots, Z_k)$; outcome $Y$ at the end of follow-up. For two periods: $Z_0 \to X_0$, $Z_0 \to Z_1$, $X_0 \to Z_1$, $Z_1 \to X_1$, $X_0 \to X_1$, and all of them $\to Y$.




**Treatment–confounder feedback** occurs when (1) past treatment $X_{k-1}$ affects a future confounder $Z_k$, and (2) $Z_k$ affects future treatment $X_k$. Example (HIV): antiretroviral therapy (ART, $X$) $\to$ CD4 count or viral load ($Z$, health biomarkers) $\to$ next ART decision $\to \cdots$. Previous treatment may also guide later decisions directly ($X_{k-1} \to X_k$). This feedback loop makes standard methods fail.

**Treatment strategies.** A strategy specifies treatment at each time. *Static*: the same for everyone regardless of covariates, e.g. "always treat" $\bar{x} = (1, \ldots, 1)$, "never treat" $\bar{x} = (0, \ldots, 0)$. *Dynamic*: treatment at time $k$ depends on observed history, $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$, e.g. "initiate ART only when CD4 drops below 500". Causal estimands contrast mean counterfactual outcomes, $\mathbb{E}[Y(\bar{x})] - \mathbb{E}[Y(\bar{x}')]$.

## 1.1 Identification assumptions

**Sequentially randomized experiments:** $X_k$ is randomly assigned with probability depending only on $(\bar{X}_{k-1}, \bar{Z}_k)$. The observational analogue: no unmeasured common causes of $X_k$ and $Y$ given $(\bar{X}_{k-1}, \bar{Z}_k)$. An unmeasured $U$ may affect $Y$ and $Z_1$ as long as it has no arrow into $X_0$ or $X_1$; then adjusting for the measured past suffices.

- **Sequential exchangeability:** $Y(\bar{x}) \perp\!\!\!\perp X_k \mid \bar{X}_{k-1}, \bar{Z}_k$ for $k = 0, \ldots, K$ (treatment as-if randomized given the observed past).
- **Consistency:** $Y(\bar{x}) = Y$ when $\bar{X} = \bar{x}$.
- **Positivity:** $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k) > 0$.

## 1.2 The fundamental dilemma

Should we adjust for $Z_1$? Adjusting removes confounding of $X_1 \to Y$ (via $Z_1$) but over-adjusts by blocking $X_0 \to Z_1 \to Y$ (and, with $U \to Z_1$, $U \to Y$, opens the collider $X_0 \to Z_1 \leftarrow U \to Y$). Not adjusting preserves $X_0$'s effect through $Z_1$ but leaves $X_1$ confounded. Traditional methods (regression, stratification, matching) cannot do both under feedback.

**G-methods** (Robins' generalized methods) are consistent under time-varying confounding: (1) the g-formula (outcome modeling); (2) IP-weighted marginal structural models (treatment modeling); (3) g-estimation of structural nested models (effect modeling). All rely on the same assumptions. Key idea: handle $Z_1$ without blocking the causal path through it.

**What you already know.** Apply the truncated factorization (Lecture 3) to $(Z_0, X_0, Z_1, X_1, Y)$: from

$$P(z_0) \cdot P(x_0 \mid z_0) \cdot P(z_1 \mid x_0, z_0) \cdot P(x_1 \mid z_1, x_0, z_0) \cdot P(y \mid x_1, z_1, x_0, z_0)$$

remove the treatment factors ($\mathrm{do}(x_0)$, $\mathrm{do}(x_1)$) and keep the rest:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_0, z_1} \mathbb{E}[Y \mid x_0, z_0, x_1, z_1]\, P(z_1 \mid x_0, z_0)\, P(z_0).$$

This is the *g-formula*, the truncated factorization in temporal order; sequential exchangeability ensures each remaining factor is estimable. What is new: why feedback breaks standard estimation, and estimation strategies (sequential weight products, MSMs, SNMs).

## 2 The g-formula

**Purpose.** Standardize over the time-varying confounder instead of conditioning on it.

**Idea** (two time points, no baseline confounder, so $X_0$ is marginally exchangeable). Instead of conditioning on $Z_1$, standardize over its distribution *given past treatment*:

$$\mathbb{E}[Y(x_0, x_1)] = \sum_{z_1} \mathbb{E}[Y \mid X_0 = x_0, Z_1 = z_1, X_1 = x_1]\, P(Z_1 = z_1 \mid X_0 = x_0),$$

weighting by $P(z_1 \mid x_0)$, not by the marginal of $Z_1$.

**Derivation.** The observed density factorizes temporally (writing $f$ and $P$ interchangeably):

$$P(y, x_1, z_1, x_0) = P(y \mid x_1, z_1, x_0) \underbrace{P(x_1 \mid z_1, x_0)}_{\text{remove}} P(z_1 \mid x_0) \underbrace{P(x_0)}_{\text{remove}}.$$

Under $\mathrm{do}(X_0 = x_0, X_1 = x_1)$ the assignment terms are replaced by the intervention, giving outcome model $\times$ confounder model as above. For $K+1$ time points,

$$\mathbb{E}[Y(\bar{x})] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{X} = \bar{x}, \bar{Z} = \bar{z}] \prod_{k=0}^{K} P(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}),$$

with $\bar{x}_{-1}, \bar{z}_{-1}$ empty so that $P(z_0 \mid \bar{x}_{-1}, \bar{z}_{-1}) = P(z_0)$.

### 2.1 Iterated conditional expectations (ICE)

The sum over $\bar{z}$ needs covariate densities $P(z_k \mid \text{past})$, hard to model for high-dimensional $Z_k$. ICE replaces the sum by nested regressions computed backward in time: each step is an outcome regression averaging out one confounder, the data doing the averaging, so no density model is needed:

$$Y \xrightarrow{\text{regress}} \hat{Q}_K \to \cdots \to \hat{Q}_0 \xrightarrow{\text{average}} \hat{\mathbb{E}}[Y(\bar{x})].$$

**Two time points.** (1) $\hat{Q}_1(z_1) = \hat{\mathbb{E}}[Y \mid x_0, z_1, x_1]$, an outcome regression at $(x_0, x_1)$; (2) $\hat{\mathbb{E}}[Y(x_0, x_1)] = \hat{\mathbb{E}}[\hat{Q}_1(Z_1) \mid X_0 = x_0]$, the average over the $X_0 = x_0$ group. The second step performs the g-formula's averaging through the empirical distribution of $Z_1$ in the selected treatment-history group.

**Three time points as variable elimination.** Eliminate $Z_2$, then $Z_1$:

$$Q_2(z_1, z_2) = \mathbb{E}[Y \mid x_0, z_1, x_1, z_2, x_2], \qquad Q_1(z_1) = \mathbb{E}[Q_2(Z_1, Z_2) \mid x_0, z_1, x_1],$$
$$\mathbb{E}[Y(x_0, x_1, x_2)] = \mathbb{E}[Q_1(Z_1) \mid X_0 = x_0].$$

For $Q_1$, create the pseudo-outcome $\tilde{Y}_i^{(2)} = \hat{Q}_2(Z_{1i}, Z_{2i})$ with treatments set to $(x_0, x_1, x_2)$, and regress it on $Z_1$ among rows with $X_0 = x_0$, $X_1 = x_1$; the data distribution averages over $Z_2$.

**General recursion.** Starting from $\hat{Q}_{K+1} = Y$, for $k = K, \ldots, 0$:

$$\hat{Q}_k = \hat{\mathbb{E}}\big[\hat{Q}_{k+1} \mid \bar{X}_k = \bar{x}_k, \bar{Z}_k\big], \qquad \hat{\mathbb{E}}[Y(\bar{x})] = \hat{\mathbb{E}}\big[\hat{Q}_0(x_0, Z_0)\big].$$

Every step is an outcome regression; ICE never models $P(z_k \mid \text{past})$, which is why it scales to many time points and high-dimensional $Z_k$.

## 3 IP weighting and marginal structural models

**Purpose.** Re-weight to a pseudo-population in which treatment is unconfounded.

**Idea.** Instead of modeling the outcome, weight each individual inversely to the probability of their observed treatment given confounders; this "breaks" the arrows $Z_k \to X_k$ (and $U$, which never points into treatment, stays harmless).

**One time point: why $1/P(X \mid Z)$.** Under exchangeability $Y(x) \perp\!\!\!\perp X \mid Z$, consistency and positivity,

$$\mathbb{E}\Big[\frac{\mathbb{1}(X = x)Y}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[\mathbb{1}(X = x)Y \mid Z]}{P(x \mid Z)}\Big] = \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z]P(x \mid Z)}{P(x \mid Z)}\Big]$$
$$= \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = \mathbb{E}_Z\big[\mathbb{E}[Y(x) \mid Z]\big] = \mathbb{E}[Y(x)].$$

Dividing by $P(x \mid Z)$ up-weights under-represented $(X, Z)$ combinations, building a pseudo-population with $X \perp\!\!\!\perp Z$: the arrow $Z \to X$ is gone and the crude association becomes causal.

**Two time points** (no baseline confounder), for a fixed regime $(x_0, x_1)$:

$$W^X(x_0, x_1) = \frac{\mathbb{1}(X_0 = x_0)}{P(x_0)} \cdot \frac{\mathbb{1}(X_1 = x_1)}{P(x_1 \mid x_0, Z_1)},$$

$$\mathbb{E}[W^X(x_0, x_1)Y] = \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] \frac{P(x_0, z_1, x_1)}{P(x_0)P(x_1 \mid x_0, z_1)}$$
$$= \sum_{z_1} \mathbb{E}[Y \mid x_0, z_1, x_1] P(z_1 \mid x_0) = \mathbb{E}[Y(x_0, x_1)].$$

The first factor undoes the observed assignment of $X_0$, the second that of $X_1$ given history $(X_0, Z_1)$. With baseline $Z_0$, replace $P(X_0 = x_0)$ by $P(X_0 = x_0 \mid Z_0)$.

**The weight product.** Confounding $Z_k \to X_k$ enters at every step, so divide by $P(X_k \mid \text{past})$ at every $k$:

$$W^X = \prod_{k=0}^{K} \frac{1}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)} \qquad \text{(non-stabilized)}.$$

**Stabilized weights.** The mean of $W^X$ blows up: for binary treatments,

$$\mathbb{E}[W^X] = \sum_{\bar{x}, \bar{z}} \frac{P(\bar{x}, \bar{z})}{\prod_k P(x_k \mid \text{past})} = \sum_{\bar{x}, \bar{z}} \prod_k P(z_k \mid \text{past}) = \sum_{\bar{x}} 1 = 2^{K+1},$$

so the pseudo-population is $2^{K+1}$ times larger, and rare treatment cells give extreme weights and high variance. Fix: put $P(X_k \mid \bar{X}_{k-1})$ (treatment given past treatment only) in the numerator,

$$SW^X = \prod_{k=0}^{K} \frac{P(X_k \mid \bar{X}_{k-1})}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

The numerator uses treatment history only, so it does not reintroduce $Z \to X$ confounding (the estimator remains valid), and $\mathbb{E}[SW^X] = 1$: the pseudo-population has the data's size, with weights centered at 1 and much smaller variance.

**Proof (one time point).** With $SW = P(X)/P(X \mid Z)$,

$$\begin{aligned}\mathbb{E}\Big[\frac{P(x)}{P(x \mid Z)} \mathbb{1}(X = x) Y\Big] &= P(x)\, \mathbb{E}_Z\Big[\frac{\mathbb{E}[Y \mid X = x, Z] P(x \mid Z)}{P(x \mid Z)}\Big] \\ &= P(x)\, \mathbb{E}_Z\big[\mathbb{E}[Y \mid X = x, Z]\big] = P(x)\, \mathbb{E}[Y(x)].\end{aligned}$$

With $Y \equiv 1$, $\mathbb{E}[SW\, \mathbb{1}(X = x)] = P(x)$, so $\mathbb{E}[SW] = \sum_x P(x) = 1$ and $\mathbb{E}[SW\, \mathbb{1}(X = x)Y]/\mathbb{E}[SW\, \mathbb{1}(X = x)] = \mathbb{E}[Y(x)]$. The extra factor $P(x)$ merely rescales the $X = x$ arm to its marginal size, so the pseudo-population is not doubled; it cancels in the ratio (and in an MSM), leaving the causal target unchanged and only shrinking variance.

**Proof (general).** Let $r_k(x_k \mid \bar{x}_{k-1}) \equiv P(X_k = x_k \mid \bar{X}_{k-1} = \bar{x}_{k-1})$. The observed law factorizes as $p(\bar{z}, \bar{x}, y) = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1}) p(x_k \mid \bar{x}_{k-1}, \bar{z}_k) \cdot p(y \mid \bar{x}, \bar{z})$. For $SW^X = \prod_k r_k(X_k \mid \bar{X}_{k-1})/p(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$, the weighted population has density

$$p_{SW}(\bar{z}, \bar{x}, y) = p(\bar{z}, \bar{x}, y)\, SW^X = \prod_{k=0}^{K} p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r_k(x_k \mid \bar{x}_{k-1}) \cdot p(y \mid \bar{x}, \bar{z}),$$

a proper density (so $\mathbb{E}[SW^X] = 1$) in which treatment depends only on past treatment: the $Z_k \to X_k$ arrows are removed. For a static regime $\bar{x}$ let $r(\bar{x}) = \prod_k r_k(x_k \mid \bar{x}_{k-1})$. Then

$$\mathbb{E}\big[SW^X \mathbb{1}(\bar{X} = \bar{x}) Y\big] = \sum_{\bar{z}} \mathbb{E}[Y \mid \bar{x}, \bar{z}] \prod_k p(z_k \mid \bar{x}_{k-1}, \bar{z}_{k-1})\, r(\bar{x}) = r(\bar{x})\, \mathbb{E}[Y(\bar{x})]$$

by the g-formula, and $\mathbb{E}[SW^X \mathbb{1}(\bar{X} = \bar{x})] = r(\bar{x})$, so $\mathbb{E}[SW^X \mathbb{1}(\bar{X} = \bar{x})Y]/\mathbb{E}[SW^X \mathbb{1}(\bar{X} = \bar{x})] = \mathbb{E}[Y(\bar{x})]$. Stabilization changes the marginal treatment distribution in the pseudo-population, not the causal target.

### 3.1 Marginal structural models

An MSM specifies the relation between treatment strategy and counterfactual mean, e.g.

$$\mathbb{E}[Y(x_0, x_1)] = \beta_0 + \beta_1 x_0 + \beta_2 x_1 + \beta_3 x_0 x_1.$$

It is a model for potential outcomes, not for observed data, estimated by weighted regression in the IP-weighted pseudo-population. More complex MSMs: saturated (one parameter per strategy); dose–response $\mathbb{E}[Y(\bar{x})] = \beta_0 + \beta_1 \mathrm{cum}(\bar{x})$ with $\mathrm{cum}(\bar{x}) = \sum_k x_k$; conditional MSMs including baseline covariates.

## 4 G-estimation of structural nested models

**Purpose.** Model and remove the history-specific effect of each decision.

**The blip.** An MSM models $\mathbb{E}[Y(\bar{x})]$ marginally (averaged over history). A structural nested mean model (SNMM) targets the local, history-specific effect of each decision, the *blip*: a contrast of potential outcomes with the future held at the reference (0):

$$\gamma_1(x_0, z_1) = \mathbb{E}\big[Y(x_0, 1) - Y(x_0, 0) \mid X_0 = x_0, Z_1 = z_1\big] \text{ (time-1 blip, given history)},$$

$$\gamma_0(x_0) = \mathbb{E}\big[Y(x_0, 0) - Y(0, 0)\big] \text{ (time-0 blip)}.$$

Blips accumulate: in a three-decision example from "never" (000) to "always" (111), $\mathbb{E}[Y(\bar{x})]$ climbs $000 \to 100 \to 110 \to 111$ by steps $+\gamma_0, +\gamma_1, +\gamma_2$; the MSM is the whole climb, the SNMM the history-specific steps.

**G-estimation: find the blip by removing it.** If the late blip were known, subtracting it would rebuild each patient's outcome had they not taken the late treatment: $Y_{\text{removed}} = Y - \psi_1 X_1$ (a candidate blip). G-estimation solves for the $\psi_1$ that makes $Y_{\text{removed}}$ balanced across $X_1$ given history: an estimating equation, not an outcome regression. Two handles, one target: IPW models the treatment (re-weighting to break confounding); SNMM models the effect (the blip) and removes it. Both recover $\mathbb{E}[Y(\bar{x})]$.

**In practice: one time point.** With an additive, history-homogeneous blip $\gamma(z; \psi) = \psi x$, blipping down gives $H(\psi) = Y - \psi X$, the candidate $Y(0)$. At the true $\psi$, consistency and the blip make $H(\psi) = Y(0)$, and exchangeability gives $Y(0) \perp\!\!\!\perp X \mid Z$, so $H(\psi)$ must be uncorrelated with $X$ given $Z$. With a fitted propensity $\hat{e}(Z) = \hat{P}(X = 1 \mid Z)$, solve

$$U(\psi) = \sum_i \big(X_i - \hat{e}(Z_i)\big) \underbrace{(Y_i - \psi X_i)}_{H_i(\psi)} = 0 \quad \Longrightarrow \quad \hat{\psi} = \frac{\sum_i (X_i - \hat{e}_i) Y_i}{\sum_i (X_i - \hat{e}_i) X_i},$$

an IV-type ratio in which the propensity residual $X - \hat{e}$ plays the role of an instrument. Recipe: (1) fit $\hat{e}(Z)$ (e.g. logistic); (2) form residuals $X_i - \hat{e}_i$; (3) plug in. It is consistent if $\hat{e}$ is correct; adding an outcome term $\mathbb{E}[H(\psi) \mid Z]$ makes it doubly robust. With effect modification, e.g. $\gamma(z; \psi) = (\psi_0 + \psi_1 z)x$, $H$ uses the $z$-specific blip and the instrument becomes the vector $(1, z)(X - \hat{e})$.

**Two time points.** Order $X_0 \to Z_1 \to X_1 \to Y$ with history-homogeneous blips $\gamma_1 = \psi_1 x_1$, $\gamma_0 = \psi_0 x_0$ (history enters only through the propensities). Fit the latest blip first, then work backward.

1. Remove the time-1 blip: $H_1(\psi_1) = Y - \psi_1 X_1$ is the candidate $Y(x_0, 0)$, and exchangeability gives $Y(x_0, 0) \perp\!\!\!\perp X_1 \mid (X_0, Z_1)$. With $\hat{e}_1 = \hat{P}(X_1 = 1 \mid X_0, Z_1)$, solving $\sum_i (X_{1i} - \hat{e}_{1i}) H_{1i}(\psi_1) = 0$ gives
$$\hat{\psi}_1 = \frac{\sum_i (X_{1i} - \hat{e}_{1i}) Y_i}{\sum_i (X_{1i} - \hat{e}_{1i}) X_{1i}}.$$

2. Remove the time-0 blip from the blipped-down outcome: $H_0 = H_1(\hat{\psi}_1) - \psi_0 X_0 = Y - \hat{\psi}_1 X_1 - \psi_0 X_0$, the candidate $Y(0, 0)$. With $\hat{e}_0 = \hat{P}(X_0 = 1 \mid Z_0)$,
$$\hat{\psi}_0 = \frac{\sum_i (X_{0i} - \hat{e}_{0i}) H_{1i}(\hat{\psi}_1)}{\sum_i (X_{0i} - \hat{e}_{0i}) X_{0i}}.$$

The same one-step equation is applied backward with history-indexed propensity residuals $X_k - \hat{e}_k$. Telescoping, $\mathbb{E}[Y(1, 1)] - \mathbb{E}[Y(0, 0)] = \hat{\psi}_0 + \hat{\psi}_1$. Each step becomes doubly robust by adding an outcome term $\mathbb{E}[H_k \mid \text{history}]$.

## 5 Doubly robust estimation

**Purpose.** Combine outcome and treatment models for a second chance at consistency.

The g-formula needs a correct outcome model $\mathbb{E}[Y \mid \bar{X}, \bar{Z}]$; IP weighting needs correct treatment models $P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)$; misspecification biases either. Doubly robust (augmented IPW, AIPW) estimators are consistent if either model is correct.

**Time-fixed review.** With outcome model $\hat{m}(x,z) = \hat{\mathbb{E}}[Y \mid X = x, Z = z]$ and propensity $\hat{\pi}(z) = \hat{P}(X = 1 \mid Z = z)$,

$$\hat{\tau}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\hat{m}(1,Z_i) - \hat{m}(0,Z_i) + \frac{X_i\{Y_i - \hat{m}(1,Z_i)\}}{\hat{\pi}(Z_i)} - \frac{(1-X_i)\{Y_i - \hat{m}(0,Z_i)\}}{1-\hat{\pi}(Z_i)}\right].$$

If $\hat{m}$ is correct, the IPW residual terms have mean zero; if $\hat{\pi}$ is correct, the residuals correct the bias of $\hat{m}$.

**Time-varying extension.** Fit treatment models $\hat{\pi}_k$ at each time and outcome models backward from $K$ to 0 (ICE-style); the estimator augments IPW with outcome predictions at each step and is consistent if, at each time point, either the treatment or the outcome model is correct. The full sequential AIPW involves iterated regressions and cumulative weights (Bang and Robins 2005). Modern variants: TMLE, sequential regression with cross-fitting.

## 6 Practical considerations

**Purpose.** Dynamic strategies and positivity.

**Dynamic strategies.** All three g-methods generalize to $g = [g_0(Z_0), g_1(\bar{X}_0, Z_1), \ldots, g_K(\bar{X}_{K-1}, Z_K)]$ with $g_k(\cdot) \in \{0,1\}$. G-formula: simulate forward, drawing $Z_k$ and setting $X_k = g_k(\cdot)$. IPW: weight individuals who happened to follow $g$,

$$W^g = \prod_{k=0}^{K} \frac{\mathbb{1}\big(X_k = g_k(\bar{X}_{k-1}, Z_k)\big)}{P(X_k \mid \bar{X}_{k-1}, \bar{Z}_k)}.$$

G-estimation: SNMMs naturally accommodate dynamic regimes via effect modification by $Z_k$.

**Positivity violations.** With $K$ binary treatment times there are $2^K$ treatment histories $(2, 4, 8, \ldots)$, so cells become sparse (near-positivity violations), IP weights extreme, and estimates unstable. Remedies: stabilized weights, weight truncation, parametric models.

## 7 Comparison of g-methods

**Purpose.** Summarize what each method models, targets and risks.

| | G-formula | IPW / MSM | G-estimation / SNM |
|---|---|---|---|
| Key model | outcome + confounder dist. | treatment assignment | treatment + blip function |
| Target | $\mathbb{E}[Y(\bar{x})]$ (marginal) | $\mathbb{E}[Y(\bar{x})]$ (marginal) | conditional blip effects |
| Effect modification by $Z_k$ | outcome-model interactions | not directly | directly modeled |
| Extreme weights? | no | yes | no |
| G-null paradox? | yes (parametric) | no | no |
| DR version? | yes | yes (AIPW) | yes |

All three are consistent under correct specification and the same identifiability conditions. (The g-null paradox: with parametric models for both outcome and time-varying confounders, the parametric g-formula can reject the null of no effect even when it is true, because the models cannot all be correct under the null.)

**Estimand, model, estimator.**

| Method | Estimand (what) | Model (what must be right) | Estimator (how) |
|---|---|---|---|
| g-formula | $\mathbb{E}[Y(\bar{x})]$ | outcome (+ confounders) | standardize / ICE |
| IPW | $\mathbb{E}[Y(\bar{x})]$ | treatment | weighting |
| MSM | $\mathbb{E}[Y(\bar{x})]$, marginal | *marginal form* + treatment | weighted regression |
| SNMM / g-est. | blip $\gamma_k$ (history-specific) | *blip form* + treatment | g-estimation (balance) |
| AIPW / DR | $\mathbb{E}[Y(\bar{x})]$ | outcome or treatment | augmented IPW |

Two kinds of "model": *structural* (an assumption on the estimand: MSM form, SNMM blip; saturated means no assumption, italic above) and *nuisance* (a working model of the observed data, outcome or propensity). So MSM/SNMM carry an extra structural assumption that the g-formula and IPW do not. Mind the level: MSM and SNMM are models placed on an estimand; IPW, g-estimation and AIPW are estimators (MSMs are fitted by IPW, SNMMs by g-estimation). SNMM is the outlier, targeting a different estimand, the history-specific blip, which enables time-varying effect modification and optimal dynamic regimes.

**The big picture.** Treatment–confounder feedback ($X_0 \to Z_1 \to X_1 \to \cdots$) defeats standard methods. The g-formula standardizes over time-varying $Z$; IPW/MSM re-weights to remove $Z \to X$ arrows; g-estimation/SNM subtracts blip effects backward in time; doubly robust AIPW combines outcome and treatment models.

**Key takeaways.**

1. Time-varying treatments require extending the causal framework to multiple time points and treatment strategies.
2. When a time-varying confounder is affected by prior treatment, standard regression, stratification and matching all fail.
3. The g-methods handle this under sequential exchangeability, consistency and positivity.
4. Doubly robust methods add protection against misspecification.
5. In practice choose by problem structure, available models, and sensitivity to positivity violations.

# 8 Source coverage

The source deck has 85 physical pages (34 numbered slides plus appendix).

| Topic | Physical pages |
|---|---|
| Time-varying treatments, feedback, strategies, assumptions, dilemma | 1–20 |
| G-formula and ICE | 21–41 |
| IP weighting, stabilization, MSMs | 42–55 |
| Blips and g-estimation | 56–61 |
| Doubly robust estimation | 62–68 |
| Dynamic strategies, positivity, comparisons, takeaways | 69–77 |
| Appendix: stabilized-weight proofs, g-estimation in practice, references | 78–85 |